FINANCIAL TRANSACTIONS AND LOANS DASHBOARD

Using Microsoft Power BI

PROJECT REPORT

Submitted By:

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Tool Used: Microsoft Power BI Desktop

1. Introduction

Power BI is a business intelligence and data visualization tool developed by Microsoft. It helps me organize data and present information through interactive dashboards, charts and summary cards.

For this project, I created a Financial Transactions and Loans Dashboard using Microsoft Power BI. I used account-type, transaction, monthly and loan-related information to review financial activity and compare the values shown in the dashboard.

The dashboard focuses on the distribution of accounts by account type, transaction activity by month, total loans by status and transactions by type. I also created a Transaction Tooltip page to display key transaction measures.

2. Objectives of the Project

The main objectives of my project are:

- To compare the number of accounts across different account types.
- To analyse how transaction counts change from month to month.
- To present loan information by loan status.
- To compare transactions across different transaction types.
- To summarize important transaction measures using KPI cards.
- To create a tooltip page that displays transaction value and transaction count.

3. Dataset Description

I used financial data containing account, transaction and loan information. The visible dashboard fields include Account Type, Total Accounts, Month, Total Transactions, Loan Status and Transaction Type. The tooltip page displays Total Transaction Value and Total Transactions.

The main fields and measures visible in my dashboard are:

Field / Measure	Description
Account Type	Groups accounts into Business, Youth, Savings, Payroll and Checking.
Total Accounts	Shows the account count for each account type.
Month	Used to organize transaction activity by month.
Total Transactions	Displays the number of transactions.
Loan Status	Used to group loan records according to their status.
Transaction Type	Used to categorize transactions by type.
Total Transaction Value	Displays the combined monetary value of transactions on the tooltip page.

4. Data Preparation

I organized the financial information so that I could compare accounts by account type and transactions by month. I also used loan status and transaction type as categories for the related dashboard visuals.

For the numerical summaries, account counts, transaction counts and transaction values need to be treated as numeric fields. Month values need to be arranged in calendar order so that the monthly transaction chart follows the correct sequence.

5. Summary Cards and Transaction Measures

The main dashboard includes a card displaying 1053. The label for this card is not visible in the provided view, so I have not assigned a measure name to that value.

The separate Transaction Tooltip page displays two summary cards:

- **Total Transaction Value:** 122.69M.
- **Total Transactions:** 49K.

These cards provide a quick summary of the monetary value and number of transactions represented by the tooltip context.

6. Dashboard Components

I created different visuals to review account distribution, monthly transaction activity, loan status and transaction categories.

6.1 Accounts by Account Type

I used a column chart to compare the total number of accounts across five account types: Business, Youth, Savings, Payroll and Checking. Business has the highest visible account count at approximately 350. Youth and Savings follow, while Payroll and Checking are slightly lower, at around 305 each.

Purpose: To compare account distribution across account types and identify which account categories have more accounts.

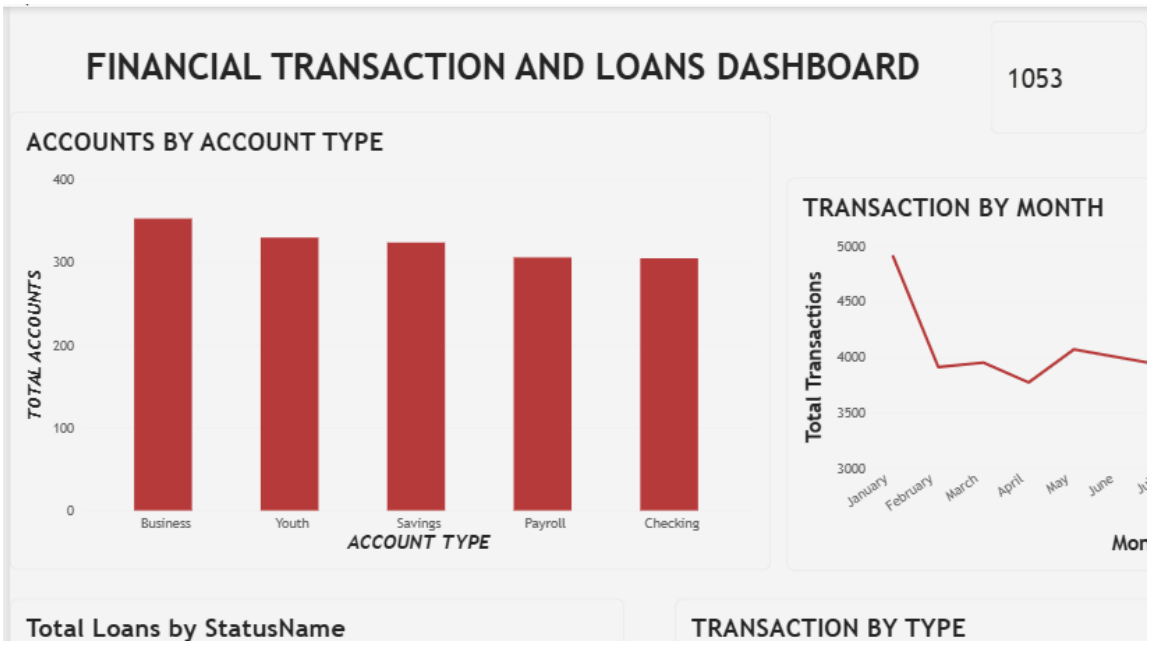


Figure 1: Account counts by type and monthly transaction trend.

6.2 Transaction by Month

I used a line chart to display total transactions by month. January begins at approximately 4,900 transactions. The value falls to around 3,900 in February, rises slightly in March, and drops again in April. It increases in May and then decreases slightly in June. The remaining months continue beyond the visible portion of the chart.

Purpose: To identify monthly changes in transaction activity and compare periods with higher or lower transaction counts.

6.3 Total Loans by Status

I included a visual titled 'Total Loans by StatusName' to group loan records by their status. This visual is part of the dashboard layout, but its plotted values are below the visible area in the supplied dashboard view. I therefore use it to present the loan-status breakdown without assigning specific counts or identifying a leading status.

Purpose: To compare the number of loans across the available loan statuses.

6.4 Transaction by Type

I included a visual titled 'Transaction by Type' to organize transaction activity by transaction category. The chart values are not visible in the supplied view, so I do not infer which transaction type has the highest count.

Purpose: To compare transaction activity across the different transaction types.

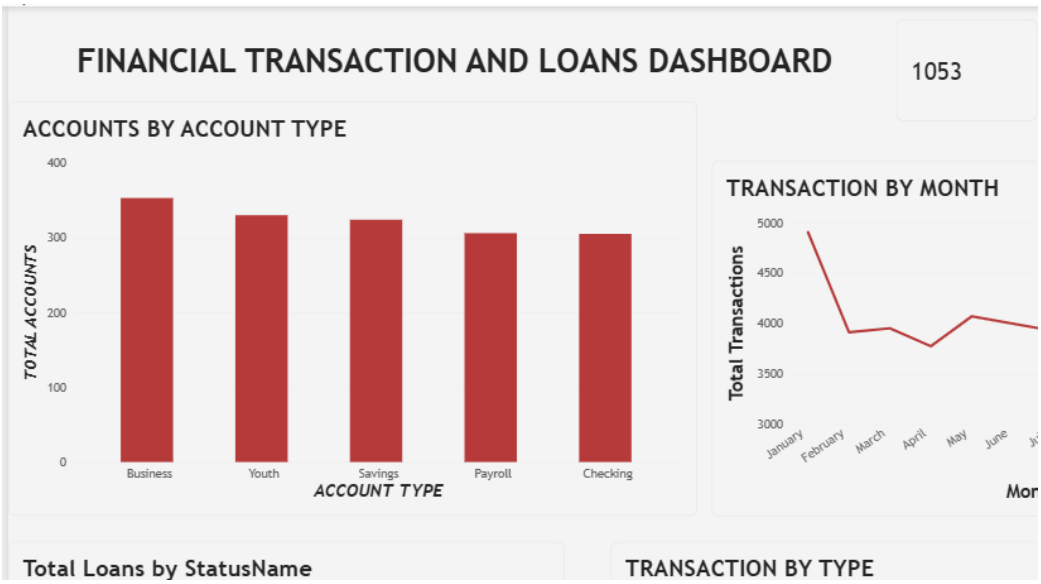


Figure 2: Main Financial Transactions and Loans dashboard view.

7. Transaction Tooltip

I created a separate Transaction Tooltip page containing two KPI cards. The first card displays Total Transaction Value of 122.69M, and the second card displays Total Transactions of 49K.

When the tooltip page is assigned to a related visual in Power BI, it can display these summary measures when I hover over a data point. This provides additional transaction information without moving away from the main dashboard.

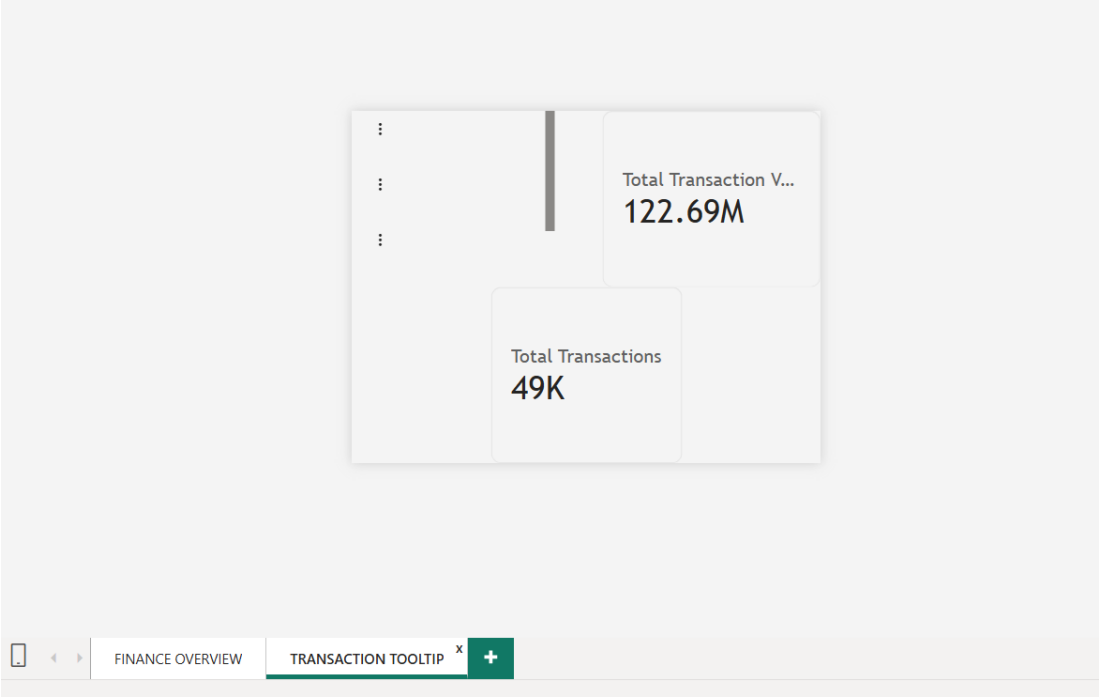


Figure 3: Transaction Tooltip page with total transaction value and total transactions.

8. Key Findings

From the values and chart patterns visible in my dashboard, I identified the following findings:

- The dashboard displays a card value of 1053, although its measure label is not visible.
- Business accounts have the highest visible account count among the five account types.
- The visible monthly transaction trend starts near 4,900 in January and falls to around 3,900 in February.
- Transaction activity reaches a lower level around April before rising in May.
- The dashboard includes separate visuals for loan status and transaction type.
- The Transaction Tooltip displays a total transaction value of 122.69M and total transactions of 49K.

9. Importance of the Dashboard

- **Account Analysis:** The account-type chart helps me compare the distribution of accounts.
- **Monthly Monitoring:** The line chart helps me review changes in transaction activity over time.
- **Loan Review:** The loan-status visual provides a way to compare loans by status.
- **Transaction Analysis:** The transaction-type visual organizes transactions into categories.
- **Quick Summary:** Tooltip cards display the total transaction value and transaction count.

10. Conclusion

Through this project, I learned how to present financial account, transaction and loan information in Microsoft Power BI. I created an account-type chart to compare account counts and a monthly line chart to review changes in transaction activity. I also included visuals for loan status and transaction type to support further financial analysis.

The Transaction Tooltip page adds a compact view of the total transaction value and total number of transactions. Together, these report elements help me review different aspects of financial activity in one dashboard.

Overall, this project improved my understanding of financial data visualization, category-based comparisons, monthly trend analysis, KPI cards and tooltip design using Power BI.

11. References

- Microsoft Power BI Desktop.
- Financial transactions and loans dataset used for the project.
- Microsoft Power BI documentation.